

Radioactive Waste Classification, Non Destructive Assay and the Role of Digital Twins

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Radioactive waste management requires precise and accurate classification of waste, evidence based estimates of radionuclide inventories and sound strategies for long term storage, disposal and repurposing. In the United Kingdom a new policy framework emphasises the use of the waste hierarchy, lifecycle consideration and a graded, risk informed approach to regulation. Alongside this, digital twins that couple detailed physical models to live data streams are emerging as tools for monitoring and decision support in nuclear facilities. This report links these developments using a laboratory gamma cube phantom that represents a low level waste package. High resolution gamma spectra from a Canberra HPGe detector are combined with Cs-137 transmission measurements from a NaI(Tl) detector in a Python workflow that performs radionuclide identification from a set of known emission energies, extracts activity, efficiency and resolution, and carries out Monte Carlo based attenuation modelling in a $3 \times 3 \times 3$ grid of Cu, Al and PMMA voxels. The Cs-137 transmission data are used with the Monte Carlo model to reconstruct the internal material layout and to cross check the aluminium density against reference data, while the HPGe cube spectra are used to fit a four nuclide activity model dominated by Eu-152 and Y-88 with smaller Ba-133 and Cs-137 contributions. A separate six face analysis then uses the reconstructed attenuation map and normalised count patterns to localise the principal Eu-152, Y-88 and Ba-133 sources to restricted regions of the cube volume. Using the Eu-152 calibration source, the total Eu-152 activity in the cube is estimated to be of order 10^2 Bq giving an activity concentration many orders of magnitude below UK low level waste limits so that the phantom represents straightforward LLW or VLLW material. The integrated modelling and measurement chain is interpreted as a small scale prototype for digital twins that support radioactive waste classification, disposal strategy and repurposing routes.

I. INTRODUCTION

Radioactive waste is generated by nuclear power production as well as by medical and industrial applications of radionuclides and their use in research laboratories [5]. The decommissioning of ageing reactors and experimental facilities generate large volumes of varying, often poorly documented, material. Management of these wastes must keep doses to workers, the public and the environment as low as reasonably possible while making use of limited disposal space and maximising the opportunity to recycle waste.

The recent United Kingdom policy framework on managing radioactive substances and nuclear decommissioning practices provides a unified approach across England, Scotland, Wales and Northern Ireland, promoting the application of the waste hierarchy, continuous lifecycle considerations and a graded, risk informed approach where regulatory effort is proportional to hazard and risk [1]. Within this framework, solid wastes are classified into high level waste (HLW), intermediate level waste (ILW), low level waste (LLW) and very low level waste (VLLW) according to activity and heat output [2, 3]. LLW and VLLW dominate by volume, while HLW and ILW dominate by activity, so accurate but efficient classification of low activity items is crucial for making best use of limited disposal capacity.

Non-destructive assay (NDA) based on gamma ray spectrometry is a vital tool for determining the radioactive content of sealed containers [5]. In real facilities,

drums and boxes contain complex mixtures of metals, concrete, plastics and voids, which attenuate and scatter gamma rays in ways that depend on unknown and often complex internal geometries. This complicates the process of building nuclide inventories from the measured spectra. Tomographic and model based methods that include both transmission and emission measurements have been developed to address this [7–10].

Digital twins extend these ideas by maintaining continuously updated virtual models of physical systems that incorporate geometry, materials, live sensor data and predictive calculations [12]. In a waste management context digital twins can support inventory tracking, optimisation of waste routing, and safety analysis. They require well characterised links between nuclide inventories, waste geometries and detector responses.

A gamma cube phantom was used as a controlled test system. It consists of an aluminium box containing a three by three by three lattice of removable copper, aluminium and PMMA blocks, with sealed sources organised in an unknown configuration. Spectra were recorded from all faces of the cube and then used to determine which radionuclides were present, estimate their activity and locate their position, and infer the internal packing arrangement. The experiment then used these measurements to link non destructive assay methods with modelling approaches used in waste characterisation and digital twin development. The experimental set up and safety framework followed the PHYS306 Block F nuclear decommissioning laboratory script [6]. The next section

summarises the detectors, data sets and modelling framework used in the analysis, while Section III presents the key results before Sections IV and V discuss their implications for digital twins and waste routing.

II. METHODS AND BACKGROUND SCIENCE

Detectors, data sets and calibration activities

Spectra were acquired using a Canberra HPGGe-BE3825 detector coupled to a Digital Signal Analyzer from Mirion for detailed gamma line analysis. Attenuation measurements were carried out with a Mirion 802 NaI(Tl) scintillation detector (2×2 inch crystal) coupled to a Universal Digital MCA Tube Base with a certified collimated Cs-137 source. Two calibrated reference sources were used:

- Cs-137, providing a strong line at 662 keV with a simple spectrum dominated by a single photopeak.
- Eu-152, providing multiple lines spanning 120 keV to 1.4 MeV, many of which are well separated and have accurately known emission probabilities.

Eu-152 is well suited to calibration of the energy scale, resolution and efficiency over a broad range, while Cs-137 provides a clean probe of attenuation at a medium gamma energy.

Using certificate reference dates and half lives, the activities at the time of the calibration run were

$$A_{\text{Cs-137}} = (1.350 \times 10^4 \pm 6.748 \times 10^2) \text{ Bq}, \quad (1)$$

$$A_{\text{Eu-152}} = (3.953 \times 10^1 \pm 1.977 \times 10^{-2}) \text{ Bq}. \quad (2)$$

The Eu source was placed on the HPGGe detector axis at (25 ± 1) cm from the front face of the end cap using a perspex screen, and a single acquisition with a live time of (300 ± 1) s was recorded. Peak areas were obtained using narrow integration windows of order a few FWHM on either side of the centroid with sideband background subtraction with full details are given in Appendix A. The Cs source was the attenuation characteriser for NaI(Tl).

Two data sets were analysed:

1. **Gamma Spectroscopy** folder: Eu calibration spectrum and six spectra taken on the cube.
2. **Attenuation Data** folder: Cs spectrum taken at 27 positions corresponding to the 3×3 phantom grid on three orthogonal faces of the cube.

The measurements were carried out in the CTH radiation lab on 25/11/2025 using the bench mounted HPGGe detector and a fixed position NaI(Tl) detector.

For the six Gamma Spectroscopy cube spectra, the aluminium cube was placed on a marked position on the sample table so that its geometric centre lay on the detector axis at a distance of (25 ± 1) cm from the HPGGe end cap. The faces of the cube were labelled so that

the face directly facing the detector defined the $+x$ orientation, and the other two orthogonal faces defined $+y$ and $+z$. For each of the six positions the cube was rotated so that the required face was normal to the detector axis, then a spectrum with a live time of (1500 ± 1) s was taken. Recording spectra on all six faces rather than only three provides additional data for the six face localisation analysis and maximises the representation of all radionuclides.

For the Attenuation Data Cs-137 set, the NaI(Tl) detector was positioned at a distance of (25 ± 1) cm from the relevant face of the cube, with a lead collimator defining a narrow beam that passed through one voxel position at a time. Numbers on the cube surface marked the 3×3 grid positions. At each of the 27 positions on the three faces, the collimated beam was aligned with the centre of the voxel and a spectrum was acquired for a live time of (60 ± 1) s. The collimator and detector geometry were kept fixed so that relative count rates reflected changes in attenuation through different material paths.

For the aluminium attenuation measurements used in the density cross check, the collimated Cs-137 beam passed through the two aluminium walls of the cube, each of thickness (1.185 ± 0.001) cm. The combined effective aluminium thickness was therefore (2.370 ± 0.002) cm. A spectrum was recorded for this configuration, together with a reference spectrum taken with no aluminium in the beam. The net peak areas for the 662 keV line were then converted to count rates.

All spectra were supplied as Prospect exported CSV files. All resolution and efficiency calculations and models can be found in Appendices A and B.

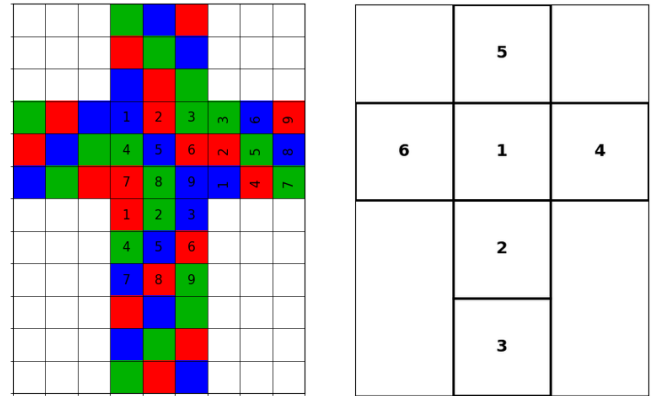


FIG. 1: Unfolded net of the gamma cube. The left panel shows the 3×3 grid of beam positions on each face, and the right panel labels the six external faces (1–6) in the orientation used for the HPGGe and NaI(Tl) measurements, where faces 1, 2 and 3 correspond to the $+x$, $+y$ and $+z$ directions.

Cube geometry and material model

The internal volume of the gamma cube was represented as a $3 \times 3 \times 3$ array of equal cubic voxels.

The voxels are indexed by coordinates (i_x, i_y, i_z) with $i_x, i_y, i_z \in \{0, 1, 2\}$, where $i_x = 0$ and $i_x = 2$ correspond to voxels on the negative and positive x faces, and similarly for y and z . If L is the inner side length of the aluminium box, the length of a voxel is $L/3$, so the centre of voxel (i_x, i_y, i_z) lies at

$$x = (i_x + \tfrac{1}{2}) \frac{L}{3}, \quad y = (i_y + \tfrac{1}{2}) \frac{L}{3}, \quad z = (i_z + \tfrac{1}{2}) \frac{L}{3}$$

Each voxel was assigned a material from $\{\text{Cu}, \text{Al}, \text{PMMA}\}$, while the external walls of the container were treated as a uniform aluminium shell with thickness that is the difference between the outer and inner dimensions. In the code, material labels are stored as a $3 \times 3 \times 3$ array and formulated as densities and energy dependent attenuation coefficients $\mu(E)$ using tabulated NIST data [15]. This voxel representation lays the foundation for both the attenuation modelling and the six face localisation. The reconstructed material layout that results from the optimisation is presented later in Fig. 2a.

For the Cs-137 attenuation measurements with the NaI(Tl) detector, nine positions were used on each of the three orthogonal faces, $+x$, $+y$ and $+z$, which are modelled as a 3×3 grid of collimated beams that traverse each column of voxels before reaching the detector.

To keep track of the mapping between file names, detector positions and voxel coordinates, an unfolded net of the cube is used and shown in 1. Each square in this net corresponds to one of the 3×3 beam positions on a given face and is labelled with its voxel indices and face identifier. The visualisation is generated directly from the same data structures that are found the Monte Carlo ray tracer and the material layout solver, so the figure and the model are consistent.

Gamma attenuation and voxel model

Gamma attenuation in matter follows the exponential law

$$I = I_0 \exp \left[- \left(\frac{\mu}{\rho} \right) x \right],$$

where I_0 and I are the incident and measured intensities, μ/ρ is the mass attenuation coefficient and x is the areal density. In a multi-material container photons travel through several regions with different μ/ρ values.

For the cube analysis the internal volume was represented as a $3 \times 3 \times 3$ lattice of cubic voxels, each labelled as Cu, Al or PMMA with corresponding density and energy dependent attenuation coefficient extracted from NIST with the external walls were treated as an additional aluminium layer.

For each detector geometry a Monte Carlo ray tracer was used to simulate many straight line photon paths from each source voxel to the detector face. For every

ray that reaches the detector the code recorded the path lengths through each voxel and through the aluminium wall, and from these lengths and the material coefficients computed an effective transmission factor. The mean transmission for the detector geometry and gamma energy is the average of these factors over all rays and models how the geometry and packing of materials reduce the measured intensity. Each geometry was simulated with an order of 10^4 photons, which keeps the Monte Carlo noise on the mean transmissions at the level of a few percent while keeping the runtime relatively low.

To determine the most probable configuration, the code compares the measured peak count rates with the predicted count rates from a trial material layout. The predicted rate is obtained by multiplying the source activity, the emission probability for each gamma line, the detector efficiency and the Monte Carlo transmission for that layout. All trial layouts are encoded as $3 \times 3 \times 3$ arrays of material labels.

For a given trial configuration the difference between measurement and model is taken as the sum of squared errors

$$\text{SSE} = \sum_i (C_i^{\text{meas}} - C_i^{\text{model}}(\text{configuration}))^2, \quad (3)$$

where the index i runs over all selected peaks and each of the detector positions. For each configuration the activities are obtained using the standard least squares solver `numpy.linalg.lstsq` from the NumPy scientific computing library, and the resulting SSE value is recorded.

To reduce the risk of becoming trapped in a local minimum, the code evaluates a wide range of candidate $3 \times 3 \times 3$ material configurations. These candidates start from an initial all aluminium layout and multiple random initial labelings of the voxels, which are then refined by updates within the `solve_layout` code. For each configuration the activities are refitted by least squares and the SSE is computed. The configuration with the smallest SSE over all configurations is then taken as the best fit. This greedy search with random restarts is easy to implement and suitable for the low size of this problem but a full scale optimiser or Markov Chain Monte Carlo could be used to improve the assay for real-world waste classification models.

Modelling assumptions The attenuation and layout model assumes several approximations:

- photons travel along straight lines from point sources at every voxel centres, with no modelling of multiple scatterings or build up
- each voxel is homogeneous and is assigned a single material and density, so smaller scale structure within a block is not assumed
- within a given energy a single effective attenuation coefficient is used for each material, neglecting small energy dependence across the window

- detector response is characterised by an energy dependent full energy peak efficiency, and any non linearities or dead time effects are factored into systematic uncertainties
- Cs-137 and Eu-152 sources are assumed to be correctly positioned within their holders to within a few millimetres so that geometric effects are dominated by the cube rather than source offsets.

These assumptions are revisited in the localisation and digital twin discussion when assessing how well the cube model can be extrapolated to real waste. The full formulation of the voxel model, the construction of the path length and transmission scalars, and the optimisation are given in Appendix A together with the uncertainty treatment.

Aluminium density from geometry and attenuation

The aluminium density of the cube walls was estimated using the measured external dimensions, internal cavity dimensions and mass of the object. If V_{out} and V_{in} are outer and inner volumes, the aluminium volume is $V_{\text{Al}} = V_{\text{out}} - V_{\text{in}}$ and the density is

$$\rho = \frac{m}{V_{\text{Al}}}.$$

Uncertainties in m and in the measured dimensions were propagated using standard error propagation.

As an independent check, Cs-137 attenuation data through the cube were fitted. For an incident intensity I_0 passing through aluminium with thickness x , the transmitted intensity is

$$I(x) = I_0 \exp(-\mu_{\text{Al}}x).$$

Fitting this model to count rates versus thickness gave an effective aluminium attenuation coefficient. Comparison with tabulated coefficients [15] provided a cross check on the density estimate. The detailed error propagation for these quantities is given in Appendix A.

Activity localisation and nuclide identification

A six face spectrum analysis was performed using the spectra recorded on the $\pm x$, $\pm y$ and $\pm z$ faces. The analysis had two stages: identify which nuclides are found within the cube spectra, and for each identified nuclide use its strongest gamma line to construct a source probability map on the $3 \times 3 \times 3$ grid.

Nuclide identification Each of the six face spectra were passed through a peak finding and line matching code. Detected peaks were compared with a catalogue of common nuclides (Eu-152, Y-88, Ba-133, Cs-137 and Na-22). For each nuclide the algorithm counted how many lines were matched and computed the average absolute energy deviation from tabulated energies.

Source localisation using the strongest line of each nuclide For each of the three significant nuclides found (Eu-152, Y-88 and Ba-133), the six face localisation was carried out using the strongest gamma line for that nuclide:

- Eu-152 $\rightarrow$ 1408 keV,
- Y-88 $\rightarrow$ 898 keV,
- Ba-133 $\rightarrow$ 356 keV.

These lines are intense, relatively well isolated and span low, medium and high energies, so they provide diverse tests of the attenuation model.

For a selected line, the net count rates I_f on the six faces were obtained by integrating over a window defined by the resolution. These six rates were collected into a vector

$$\mathbf{I}_{\text{abs}} = \begin{bmatrix} I_{xp} \\ I_{xm} \\ I_{yp} \\ I_{ym} \\ I_{zp} \\ I_{zm} \end{bmatrix}, \quad (4)$$

and normalised to give a relative pattern. In this way only the geometric pattern of intensities, not the absolute scale, defines the localisation.

Forward model from the attenuation voxel grid The reconstructed $3 \times 3 \times 3$ grid of linear attenuation coefficients μ_{ijk} from the material layout optimisation, corresponding to the best fit configuration, was computed at the energy of the chosen line and then fixed for that run. For a trial source placed at the centre of voxel (i, j, k) , straight paths were followed along $\pm x$, $\pm y$ and $\pm z$ to the corresponding cube faces. Along each path the code accumulated the line integral of μ ,

$$\tau_f(i, j, k) = \int_{\text{path}} \mu(s) ds, \quad (5)$$

giving an attenuation factor for each face. The attenuation factors were converted into intensities via

$$I_{\text{model},f}(i, j, k) \propto \exp[-\tau_f(i, j, k)], \quad (6)$$

and normalised so that the six components sum to one. This gives a relative pattern

$$\mathbf{I}_{\text{model}}(i, j, k). \quad (7)$$

Comparison with data and building the probability maps At each voxel centre the difference between measurement and model was taken as

$$\text{SSE}(i, j, k) = \|\mathbf{I}_{\text{meas}} - \mathbf{I}_{\text{model}}(i, j, k)\|^2. \quad (8)$$

The 27 values of $\text{SSE}(i, j, k)$ were then inverted to produce a probability field p_{ijk} on the grid that highlights

the most probable source voxels. The construction of the six face localisation model is given in Appendix A.

Each of the localisation maps in Figs. 2b, 2c and 2d was produced by repeating this procedure for the strongest gamma line of Eu-152, Y-88 and Ba-133 respectively. Therefore each map corresponds directly to one measured peak and shows the region of the cube most consistent with the observed six face intensity pattern for that energy.

Analysis workflow overview

The full analysis follows the stages:

1. Eu-152 HPGe calibration spectra are used to fit an approximate efficiency curve and a resolution model.
2. Cs-137 NaI(Tl) spectra in the Attenuation Data set provide relative transmissions as a function of position, which are fitted with the Monte Carlo voxel model to reconstruct the internal Cu/Al/PMMA layout and to cross check the aluminium density.
3. For the reconstructed layout, the Monte Carlo model provides mean transmission factors $\bar{A}(g, E)$ that feed into the activity matrix which links nuclide activities to measured peak count rates.
4. The six face localisation model combines the reconstructed attenuation map with normalised six face line intensities to produce source probability maps on the $3 \times 3 \times 3$ voxel grid for the strongest Eu-152, Y-88 and Ba-133 lines.

Appendix A provides the mathematical formulation of each step, and Appendix B documents the underlying spectra and calibration results. These methods define a procedure from raw data to reconstructed material layout, source localisation and activity estimates. Section III summarises the key results obtained from these methods.

III. KEY RESULTS

Nuclide identification

Across all six spectra the line matching results strongly favour Eu-152, with additional contributions from Y-88, Ba-133 and weak Na-22, as shown in Sec. II C. The summary of matched lines and mean absolute energy deviations from tabulated values is given in Table I. The large number of Eu-152 lines and their small deviations provide a confident identification of the nuclide vector for the later activity fit.

TABLE I: Summary of line matching across all six cube spectra. For each nuclide the number of matched lines and the mean absolute energy deviation from tabulated values are shown.

Nuclide	Lines matched	Mean $ \Delta E $ (keV)
Eu-152	29	1.24
Y-88	12	1.13
Ba-133	6	1.44
Na-22	2	1.18

Cube composition and fitted activities

The internal material layout of the cube was reconstructed using the Cs-137 transmission measurements from the NaI(Tl) detector. For each of the 27 beam positions on the three orthogonal faces, the 662 keV peak area was converted into a relative count rate. The Monte Carlo ray tracer then predicted the corresponding transmission factors for a given trial $3 \times 3 \times 3$ Cu/Al/PMMA layout. By comparing the measured and modelled transmission patterns and minimising the sum of squared differences over all 27 positions, the code selected the material configuration that best reproduces the Cs-137 data.

Once this Cu/Al/PMMA layout was fixed, the HPGe spectra from the six faces were used to estimate activities for a four nuclide model with Ba-133, Cs-137, Eu-152 and Y-88. For a chosen set of clean lines, the measured peak count rates were related to the unknown activities through the calibrated detector efficiency, the tabulated emission probabilities and the Monte Carlo transmission factors from the reconstructed configuration. Solving this linear system with a weighted least squares fit computes activity coefficients for the four nuclides. These coefficients include the absolute efficiency and transmission scale and are therefore given in arbitrary units, but their relative values are well constrained and can be used to infer the dominant contributors to the spectra. The best fit values and approximate one sigma uncertainties from the covariance matrix are summarised in Table II.

The large Ba-133 coefficient mainly reflects the high intrinsic efficiency for low energy lines in the setup and model. When the activities are interpreted together with the emission probabilities and the observed spectral shapes, Eu-152 and Y-88 provide the main contribution to the medium and high energy peaks, Ba-133 accounts for much of the low energy emission, also benefiting from convolution with lead x-rays and Compton scattering noise, and Cs-137 is negligible in this fit.

The material layout optimisation based on the Cs-137 transmission data found a physically consistent model. In the best fit configuration (Fig. 2a) the upper z layer is largely copper, the central layer contains a PMMA region with a mixed source dominated by Eu-152 activity, and the lower layer consists of a mixture of aluminium, copper and PMMA voxels.

The best fit weighted sum of squared errors for the Cs-

TABLE II: Fitted activity coefficients for the four nuclei identified. The A_ν are in arbitrary units that include the absolute efficiency and transmission scale. Uncertainties are one sigma values from the weighted least squares covariance matrix.

Nuclide	A_ν (arb. units)	Rel. unc.	Energy range (keV)
Ba-133	14.4 ± 33.3	2.32	300–400
Cs-137	0.521 ± 7.96	15.29	662
Eu-152	5.84 ± 14.3	2.44	800–1400
Y-88	2.87 ± 4.60	1.60	898, 1836

137 transmission fit is about 3.22 in the units used for the normalised count rates. With $N_{\text{data}} = 27$ transmission points and $N_{\text{params}} = 1$ this corresponds to a reduced chi squared of

$$\chi_{\text{red}}^2 \approx 0.12,$$

and an rms deviation between measured and modelled transmissions of about 35 %. Trial layouts in which Cu blocks are moved to unrealistic positions give SSE values that are a factor of two to three larger, which shows that the Cs-137 data carries a physically consistent description on the configuration even though the layout cannot be independently known.

Aluminium density

Using the measured mass and dimensions of the cube the bulk aluminium density was found to be

$$\rho = 2.675 \pm 0.010 \text{ g cm}^{-3},$$

to be compared with a reference value of about 2.699 g cm^{-3} . The corresponding z score of -2.4 lies within a 3σ band and indicates acceptable agreement. Fits to Cs-137 attenuation data through the aluminium box yield attenuation coefficients consistent with this density, supporting the validity of the attenuation model used in the cube analysis.

Uncertainties and error budget

For the Eu-152 calibration, counting statistics on the main peaks are at the level of a few percent or better, while the residuals of the efficiency fit (Appendix B) are at the several percent level. This implies that systematic effects from the choice of the model, source positioning and inaccurate energy calibration dominate over poissonian counting statistics, and a conservative estimate is that the absolute efficiency curve carries an uncertainty of order 5 to 10 % in most of the energy range.

For the Cs-137 transmission data, statistical uncertainties on individual 662 keV peaks are small compared with systematic uncertainties from the straight line transport

approximation and from possible small misalignments of the collimated beam on the voxel grid. These effects influence the inferred Cu/Al/PMMA pattern rather than the overall attenuation scale and are reflected in the spread of acceptable layouts with SSE values close to the minimum.

The aluminium density obtained from geometry is most sensitive to uncertainties in the external and internal dimensions rather than the mass, because the density depends on the difference between two similar volumes. The quoted $\pm 0.010 \text{ g cm}^{-3}$ uncertainty is therefore dominated by the assumed caliper errors and is consistent with the independent attenuation based density check.

Combining these effects, the total Eu-152 activity inferred for the cube is likely uncertain at the level of a factor of order two when efficiency, attenuation modelling, positioning and line selection are folded together. This level of uncertainty is small compared with the orders of magnitude between the inferred activity concentration and the regulatory boundaries between VLLW, LLW and ILW, which is why the waste classification conclusions are robust even though detailed activity values carry sizeable systematic errors.

Six face localisation

For the Eu-152 line at 1408 keV the net 1408 keV count rates from the six external faces were

$$I_{xp} = 0.096 \text{ cps}, \quad I_{xm} = 0.040 \text{ cps}, \quad (9)$$

$$I_{yp} = 0.025 \text{ cps}, \quad I_{ym} = 0.059 \text{ cps}, \quad (10)$$

$$I_{zp} = 0.082 \text{ cps}, \quad I_{zm} = 0.072 \text{ cps}. \quad (11)$$

Using the reconstructed cube attenuation model on the $3 \times 3 \times 3$ lattice, the attenuation calculation gives a minimum squared error for voxel indices

$$(i_x, i_y, i_z) = (2, 0, 2), \quad (12)$$

corresponding to approximate coordinates

$$x \approx 0.83, \quad y \approx 0.17, \quad z \approx 0.83, \quad (13)$$

with 0 at the negative face and 1 at the positive face. Thus the source of the 1408 keV line is most likely on the positive x side of the cube, closer to the negative y side and in the upper z layer. The minimum SSE is about 1.4×10^{-2} . The corresponding probability map is shown in Fig. 2b. In terms of the probability field p_{ijk} , the smallest set of voxels that contains 68% of the total probability consists of 16 voxels clustered around (2, 0, 2) in the upper layer, which provides a better measure of localisation than quoting a single best voxel.

Repeating the same procedure for the Y-88 line at 898 keV gives the normalised pattern

$$\begin{aligned} xp &= 0.085, & xm &= 0.118, & yp &= 0.251, \\ ym &= 0.075, & zp &= 0.255, & zm &= 0.216. \end{aligned} \quad (14)$$

The best fit voxel in this case is

$$(i_x, i_y, i_z) = (0, 1, 1), \quad (15)$$

with approximate coordinates

$$x \approx 0.17, \quad y \approx 0.50, \quad z \approx 0.50, \quad (16)$$

and a minimum SSE of about 2.5×10^{-2} . The associated probability map in Fig. 2c shows that the probability field is relatively flat, with a shallow maximum in the central part of the cube slightly biased toward the negative x side. For Y-88 the six face method therefore constrains the source to a broad central region rather than a sharply defined voxel. The smallest 68% probability region contains 16 voxels, which shows this as a somewhat poor localisation.

For the Ba-133 line at 356 keV, where attenuation is stronger and the straight line path model is least realistic, the measured pattern normalises to

$$\begin{aligned} xp &= 0.000, & xm &= 0.063, & yp &= 0.403, \\ ym &= 0.000, & zp &= 0.357, & zm &= 0.177. \end{aligned} \quad (17)$$

The same grid search yields a best fit at

$$(i_x, i_y, i_z) = (1, 2, 2), \quad x \approx 0.50, \quad y \approx 0.83, \quad z \approx 0.83, \quad (18)$$

with a minimum SSE of 1.2×10^{-1} that shows both the directional variation in the measured face intensities and the limitations of the simple model at low energy. Even so, the probability map in Fig. 2d peaks in the back upper region of the cube near the positive y and z faces, consistent with a Ba-133 rich source in that part of the volume. The number of voxels required to enclose 68% of the probability is 17, which again emphasises the broad and model dependent nature of the localisation at this lower energy.

The three localisation results can be summarised as

$$\text{Eu-152 (1408 keV)} : (i_x, i_y, i_z) = (2, 0, 2), \quad (19)$$

$$\text{Y-88 (898 keV)} : (i_x, i_y, i_z) = (0, 1, 1), \quad (20)$$

$$\text{Ba-133 (356 keV)} : (i_x, i_y, i_z) = (1, 2, 2). \quad (21)$$

Taken together they suggest that Eu-152 activity is concentrated toward the positive x side of the cube, while Y-88 and Ba-133 likely occupy voxels in the back upper region. Given the relatively broad probability and the simplicity of the attenuation model, these positions should be interpreted as coarse source regions rather than precise voxel indices. From a waste assay perspective this means that the method can distinguish between activity concentrated toward different faces or layers of a package, but it cannot by itself guarantee that individual voxels or small regions are empty. In a real drum this style of localisation would therefore be combined with other measurements or conservative assumptions to define bounding regions for activity.

Waste classification and activity scale

The Eu-152 calibration source has an activity of about 4×10^1 Bq at the calibration date. Comparing the Eu-152 line count rates in the cube spectra with those from the calibration run, and correcting approximately for geometry, efficiency and attenuation using the voxel model, suggests that the total Eu-152 activity in the cube is of order 10^2 Bq. In other words, the chain

Eu-152 peak rates $\rightarrow$ voxel Eu-152 activities

$\rightarrow$ total Eu-152 activity in the cube

is anchored to the calibrated source and the reconstructed attenuation pattern, and folding the fitted Eu-152 coefficient in Table II through the efficiency curve and transmission factors gives a consistent activity scale.

Even if a harsh factor of five is allowed for combined systematic uncertainties in efficiency, attenuation and spectral dampening, the activity remains well below the GBq per tonne limits that define the upper boundary of the LLW band, which are of order 10 GBq tonne $^{-1}$ for typical gamma emitters [2, 3]. For a cube mass of a few kilograms, the activity concentration is at most a few 10^{-2} MBq tonne $^{-1}$, so the final link in the chain,

total activity $\rightarrow$ activity concentration

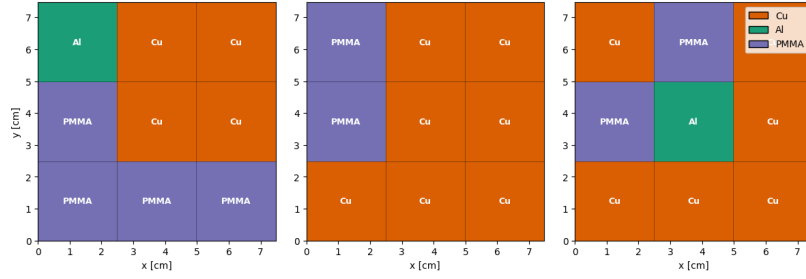
$\rightarrow$ LLW/VLLW comparison,

shows that the inferred activity concentration lies many orders of magnitude below LLW thresholds. Typical LLW upper limits lie near tens to hundreds of GBq per tonne, while VLLW limits are much lower but still far above the inferred activity concentration [2, 3]. The cube phantom can therefore be taken as material that would be classified as LLW, and probably VLLW, under UK guidance if scaled to realistic drum volumes and managed within an appropriate decay storage and clearance regime.

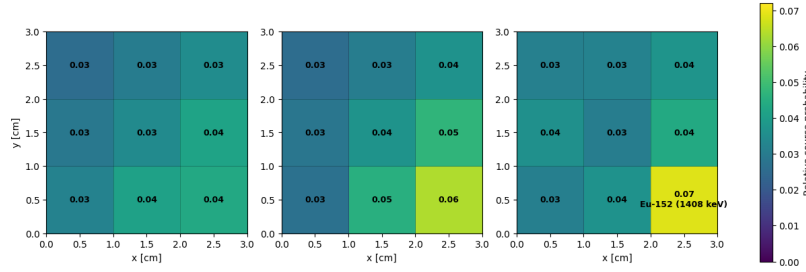
IV. DIGITAL TWINS AND MODELLING OF RADIOACTIVE WASTE

The gamma cube analysis can be viewed as a small scale prototype for a digital twin of a waste item. The cube model includes the geometry and material composition of the container (aluminium walls, internal Cu, Al and PMMA blocks and sealed sources), analogous to three dimensional CAD models of drums, boxes and stores in an industrial twin.

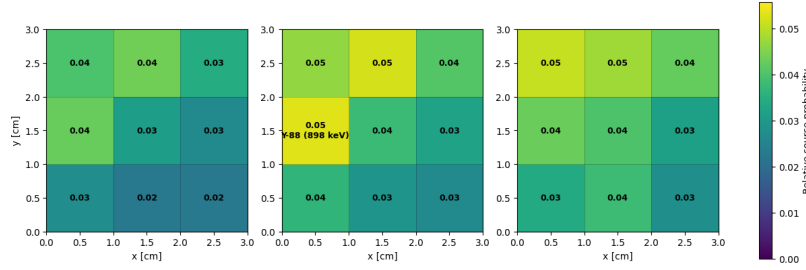
The Monte Carlo ray tracing and attenuation calculations show a forward model of detector response for given nuclide inventories and material layouts. In a full twin, more detailed radiation transport codes would be used to predict dose rates, spectra and count rates for proposed monitoring configurations or fault scenarios. The linear algebra link between measured peak rates and nuclide activities via the design matrix implies the inversion that a



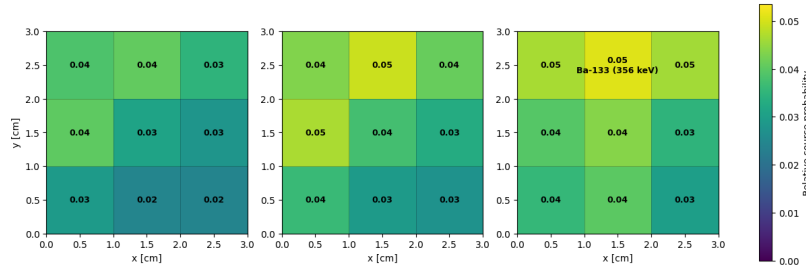
(a) Reconstructed gamma cube layout. Each panel shows one z layer of the $3 \times 3 \times 3$ interior. Voxels are labelled as Cu, Al or PMMA. The central PMMA voxel in the middle layer contains the Eu-152 dominated source along with weaker Ba-133 and Cs-137 contributions in the model.



(b) Source probability per voxel for Eu-152 (1408 keV). Relative source probability per voxel is shown for the three z layers. The probability field peaks in a voxel in the upper z layer near the positive x face and closer to the negative y side, with neighbouring voxels at comparable probability.



(c) Source probability per voxel for Y-88 (898 keV). Relative source probabilities are shown for the three z layers. The distribution is relatively broad with a shallow maximum in the central part of the cube, slightly closer to the negative x side.



(d) Source probability per voxel for Ba-133 (356 keV). The probability field is broader than for Eu-152 and Y-88 but still shows a maximum in the upper z layer near the positive y and z faces, consistent with a lower energy source in the back upper region of the cube.

FIG. 2: Gamma cube layout and source probability per voxel for the three main lines: Eu-152 (1408 keV), Y-88 (898 keV) and Ba-133 (356 keV).

digital twin would perform when updating its state from new data, and in larger systems could be extended using Bayesian inference or physics informed neural networks that approximate processor and time expensive transport calculations [10, 11].

The six face localisation demonstrates how wide but well chosen measurements can constrain source regions when combined with a physics model and an attenuation map. In a digital twin of a waste store, similar methods could support anomaly detection, identification of misplaced or degraded packages and optimisation of additional measurements. The entire workflow is implemented in a modular Python notebook that reads raw spectra, performs detector metric calculations, runs the models and produces plots and summary numbers, which is similar in structure to the software backbone of a facility scale digital twin [12].

In the current cube analysis the reduced chi squared for the Cs-137 transmission fit and the rms discrepancies between measured and modelled cube peak rates provide a first indication of how reliable the twin would be in predicting detector responses for new configurations. For the best layout the rms residuals are at the level of a few percent, which is acceptable for rough localisation and activity ratios but would be inadequate for safety critical decisions in a facility. In a full scale digital twin, discrepancies larger than an agreed threshold would trigger model updating or targeted additional measurements.

The cube twin used here omits several effects that are unavoidable in real waste packages. The internal geometry is simplified to three uniform materials in regular voxels, there is no explicit treatment of voids or water, scattering is handled only in an approximate way, and only photon emitting nuclides are considered. Real drums and boxes can contain mixed scrap with self shielding by dense items, significant voids, complex nuclide vectors including pure beta and neutron emitters, and long term evolution of activity and nuclide compositions. A facility scale digital twin would require more sophisticated geometry handling, higher level transport codes that model scattering and build up, routines for handling missing or ambiguous nuclide information, and rigorous propagation of uncertainties from measurements and models through to classification decisions.

V. DISPOSAL STRATEGY AND REPURPOSING

Link to UK waste hierarchy and classification

The UK policy framework for managing radioactive substances and nuclear decommissioning places the waste hierarchy at the centre of decision making: avoid, minimise, reuse, recycle, then dispose as a last resort [1]. It also expects duty holders to apply a proportionate, risk informed approach and to make best use of disposal capacity for different waste classes.

For an item like the gamma cube, the NDA integrated waste management strategy and UK LLW strategy suggests that the most appropriate routes are within the LLW and VLLW infrastructure rather than ILW or HLW facilities [3, 4]. The analysis above provides the key factors needed by regulators and waste managers: an estimate of activity and activity concentration for the principal nuclide (Eu-152), confirmation that only small amounts of other nuclides are present, a physical inventory of metals and polymers, and evidence that the container behaves as expected in terms of density and shielding.

Disposal plan for a gamma cube type item

An illustrative disposal plan that aligns with current UK guidance could proceed as follows.

First, the NDA and UK Radioactive Waste Inventory definitions are used to classify the cube as LLW or VLLW based on activity concentration and heat output [2, 3]. The spectrometric analysis in this work shows that the cube is well within the LLW band and likely fits the VLLW sub category if scaled to realistic drums, so a brief justification report would summarise nuclide vector, activities, uncertainties and any bounding assumptions.

Second, the waste hierarchy is applied. If the sealed Eu-152 source still has useful activity and meets encapsulation and traceability requirements, it could be removed under controlled conditions and reused as a calibration or training source. Aluminium and copper components might be sent to metal treatment facilities for decontamination and possible recycling, and PMMA blocks could be assessed for contamination and cleared for conventional disposal or reuse if activities are sufficiently low. Any decisions would follow regulatory guidance on clearance, exemption and out of scope materials.

Third, materials that cannot be cleared or recycled are conditioned as LLW or VLLW. Remaining components are size reduced and packaged in standard containers with appropriate liners and labelling, while sealed sources that are not suitable for reuse are routed via the UK sealed source disposal pathways, typically by specialist contractors working with Nuclear Waste Services. The spectrometric data and modelling enter here as part of the package record, supporting declarations of nuclide content and activity.

Finally, an appropriate disposal route is selected. Depending on the final activity concentration and the regulator approved waste route, options include disposal as VLLW at permitted landfill or dedicated VLLW facilities for lightly contaminated materials, or disposal as LLW at the national LLW repository or other facilities managed by Nuclear Waste Services for higher activity components. In a facility with a digital twin, the routing decision and associated data would be recorded so that the location and status of the waste item remain traceable through its lifecycle.

VI. CONCLUSION

This report has combined waste classification concepts, gamma spectrometric analysis and digital twin ideas using a small gamma cube phantom that represents a radioactive waste package. High resolution Eu-152 calibration data were used to determine detector efficiency and energy resolution. A tailored peak finder and line matching routine identified Eu-152 and Y-88 as the dominant nuclides in the cube, with additional contributions from Ba-133 and very small evidence for Cs-137 and Na-22. Monte Carlo based attenuation modelling in a $3 \times 3 \times 3$ voxel grid linked ray paths and material dependent attenuation coefficients to measured peak rates, and linear least squares fitting provided relative activities and a plausible internal material layout.

Independent measurements of mass, dimensions and attenuation showed that the inferred aluminium density of the cube walls is consistent with reference values. A separate six face analysis using Eu-152, Y-88 and Ba-133 lines and a simplified line integral attenuation model, built on the reconstructed μ grid, localised the sources to restricted regions near the positive x face and in the back upper part of the cube. Although the probability fields are broad and do not single out unique voxels, they are qualitatively consistent with the known configuration of an Eu-152 dominated PMMA block and a second mixed source in the upper layer. By comparing the inferred ac-

tivities with UK waste classification thresholds, the cube can be regarded as a simple example of LLW or VLLW, and an illustrative disposal plan has been sketched that follows the UK waste hierarchy and makes use of existing LLW and VLLW routes.

The analysis also suggests practical ways to improve localisation performance. On the modelling side, using energy dependent attenuation coefficients and a more realistic detector response, and treating localisation within a statistical framework that includes counting uncertainties, would reduce systematic bias and allow better defined source regions to be quoted. Experimentally, breaking the geometric symmetry by repositioning or adding detectors, introducing simple high density markers, and using a coarse to fine voxel grid focused on high probability regions would increase the information content of the six face data without excessive computational cost.

In the context of a digital twin, the gamma cube experiment shows at small scale how geometry, physical models and measurement data can be integrated into a framework that maintains a real-time representation of a waste item. Extending these concepts to operational facilities, supported by more advanced transport codes and data driven components, gives a path to better inventory control, more effective application of the waste hierarchy and confident decision making about disposal and repurposing of radioactive materials.

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Appendix A: Mathematical Derivations and Error Analysis

Mathematical derivations

This appendix summarises the mathematics used in the analysis: efficiency and resolution models, the Monte Carlo attenuation and voxel formulation, the linear algebra used for activity fitting and layout optimisation, the six face localisation model, the use of XCOM based attenuation coefficients and aluminium density measurements, and the construction of localisation probability regions.

Efficiency calibration from Eu-152

For each Eu 152 line at energy E_i the net peak area N_i is obtained by integrating counts in a window around the peak and subtracting sideband background. The count rate is

$$c_i = \frac{N_i}{t_{\text{live}}}, \quad (1)$$

where t_{live} is the live time. The absolute full energy peak efficiency at E_i is

$$\varepsilon(E_i) = \frac{c_i}{A_{\text{Eu}} P_{\gamma}(E_i)}, \quad (2)$$

with A_{Eu} the activity of the Eu 152 calibration source at the measurement date and $P_{\gamma}(E_i)$ the emission probability for that line.

A simple power law model is used for the efficiency:

$$\log_{10} \varepsilon(E) = a + b \log_{10} \left(\frac{E}{\text{keV}} \right), \quad (3)$$

which is linear in $\log_{10} E$. For measurements at energies E_i the least squares fit minimises

$$\chi^2 = \sum_i \frac{(\log_{10} \varepsilon(E_i) - a - b \log_{10}(E_i/\text{keV}))^2}{\sigma_{\log \varepsilon_i}^2}, \quad (4)$$

with $\sigma_{\log \varepsilon_i}$ obtained by propagating the uncertainties in N_i , t_{live} , A_{Eu} and $P_{\gamma}(E_i)$.

The fit parameters and goodness of fit used in the main analysis are

$$a = 2.682, \quad b = -1.014, \quad \chi^2/\text{NDF} = 45.2/4 \approx 11.3. \quad (5)$$

Efficiency calibration from Cs-137

For the Cs 137 no attenuation calibration run a single NaI(Tl) spectrum is used to estimate the intrinsic full energy peak efficiency at 661.66 keV. A Gaussian peak fit within an energy window $[E_{\text{min}}, E_{\text{max}}]$ gives a net peak area N_{Cs} and a live time t_{Cs} , so that

$$c_{\text{Cs}} = \frac{N_{\text{Cs}}}{t_{\text{Cs}}} \quad (6)$$

is the net count rate in the 661.6 keV peak. If A_{Cs} is the activity of the Cs 137 source at the measurement date and P_{γ}^{Cs} is the emission probability of the 661.6 keV line, the intrinsic efficiency at 661.6 keV is

$$\varepsilon_{\text{Cs}} = \varepsilon(661.6 \text{ keV}) = \frac{c_{\text{Cs}}}{A_{\text{Cs}} P_{\gamma}^{\text{Cs}}}. \quad (7)$$

In the model used in the code this intrinsic efficiency is treated as energy dependent over the range of interest. Energy dependent attenuation by the aluminium walls is handled separately using the Monte Carlo path lengths and XCOM based $\mu(E)$ for aluminium.

Energy resolution

For each Eu 152 line the peak is fitted with a Gaussian on a linear background. The standard deviation in energy units is σ_E and the full width at half maximum is

$$\text{FWHM}(E_i) = 2.3548 \sigma_E. \quad (8)$$

The squared resolution is modelled as linear in energy,

$$\text{FWHM}^2(E) = a_{\text{res}} + b_{\text{res}} E, \quad (9)$$

where a_{res} represents an energy independent term (for example electronic noise) and $b_{\text{res}} E$ represents a term that grows with energy. The parameters a_{res} and b_{res} are obtained from a straight line fit to the measured $\text{FWHM}^2(E_i)$ values.

From the Python fit the values obtained are

$$a_{\text{res}} = 1.166 \text{ keV}^2, \quad b_{\text{res}} = 3.1 \times 10^{-3} \text{ keV}, \quad \chi^2/\text{NDF} = 11.9/4 \approx 3.0, \quad (10)$$

as summarised in Appendix B. The fitted resolution model is used to define integration windows and to check whether peaks are expected to be separable at given energies.

XCOM based attenuation coefficients and interpolation

The NIST XCOM database provides tabulated mass attenuation coefficients $\mu/\rho(E_k)$ at energies E_k (in MeV). For a material of density ρ the linear attenuation coefficient is

$$\mu(E) = \rho \frac{\mu}{\rho}(E). \quad (11)$$

In the analysis the tabulated values are interpolated in log energy and log attenuation space. If

$$x_k = \ln E_k, \quad y_k = \ln \left[\left(\frac{\mu}{\rho} \right) (E_k) \right], \quad (12)$$

then for a given energy E with $x = \ln E$ between x_k and x_{k+1} the code uses

$$y(x) \approx (1 - t) y_k + t y_{k+1}, \quad t = \frac{x - x_k}{x_{k+1} - x_k}. \quad (13)$$

This gives

$$\frac{\mu}{\rho}(E) \approx \exp[(1 - t) y_k + t y_{k+1}], \quad (14)$$

and therefore

$$\mu(E) \approx \rho \exp[(1 - t) y_k + t y_{k+1}]. \quad (15)$$

For energies outside the tabulated range E is clipped to the range spanned by the E_k before interpolation.

Monte Carlo attenuation model, voxel representation and geometry

Gamma attenuation for a homogeneous slab obeys

$$I = I_0 \exp \left[- \left(\frac{\mu}{\rho} \right) x \right], \quad (16)$$

where I_0 and I are incident and transmitted intensities, μ/ρ is the mass attenuation coefficient and x is the areal density. For a material of density ρ and thickness L we have $x = \rho L$ and the linear coefficient is $\mu(E) = (\mu/\rho)(E) \rho$.

For the cube the interior is divided into $3 \times 3 \times 3$ cubic voxels of edge length c . Voxel j is assigned a material label $m_j \in \{\text{Cu, Al, PMMA}\}$, with density ρ_{m_j} and linear attenuation coefficient $\mu_{m_j}(E)$ at energy E . The external walls are aluminium with coefficient $\mu_{\text{Al}}(E)$.

The physical cube occupies

$$[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}], \quad (17)$$

with $x_{\max} - x_{\min} = 3c$ and same in y and z . The aluminium walls extend this region by a fixed thickness t_{wall} and the detector face is a circular disk of radius R_{det} placed a distance d_{off} outside the wall.

For each detector geometry g the Monte Carlo ray tracer:

- Samples a source position $\mathbf{x}_0$ uniformly in the chosen source voxel.
- Samples an isotropic direction $\hat{\mathbf{d}}$ with

$$\cos \theta \sim \mathcal{U}(-1, 1), \quad \phi \sim \mathcal{U}(0, 2\pi), \quad (18)$$

and

$$\hat{\mathbf{d}} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta). \quad (19)$$

- Intersects the ray $\mathbf{x}(t) = \mathbf{x}_0 + t\hat{\mathbf{d}}$ with the detector plane on the chosen face and checks whether the impact point lies inside the detector disk of radius R_{det} .
- For rays that intersect the detector, steps the ray through the voxel grid using a three dimensional digital differential analyser (3D DDA) algorithm, adding the distance travelled inside each voxel to a running total until the ray leaves the inner cube.

For each ray i that reaches the detector the marching algorithm returns a set of segment lengths Δs_{ij} in voxels j and a total aluminium wall thickness $L_{\text{wall},i}^{(g)}$ along that ray. These quantities are collected into a path length matrix and a wall thickness vector,

$$L^{(g)} = \begin{bmatrix} \Delta s_{11} & \cdots & \Delta s_{1,27} \\ \vdots & \ddots & \vdots \\ \Delta s_{N_g,1} & \cdots & \Delta s_{N_g,27} \end{bmatrix}, \quad \mathbf{L}_{\text{wall}}^{(g)} = \begin{bmatrix} L_{\text{wall},1}^{(g)} \\ \vdots \\ L_{\text{wall},N_g}^{(g)} \end{bmatrix},$$

where N_g is the number of rays simulated for geometry g and the 27 columns correspond to the 27 voxels.

At energy E the voxel attenuation coefficients are collected into

$$\boldsymbol{\mu}(E) = \begin{bmatrix} \mu_1(E) \\ \mu_2(E) \\ \vdots \\ \mu_{27}(E) \end{bmatrix}, \quad (20)$$

with $\mu_j(E) = \mu_{m_j}(E)$.

The total attenuation exponent for each ray is

$$\mathbf{T}^{(g)}(E) = L^{(g)}\boldsymbol{\mu}(E) + \mu_{\text{Al}}(E)\mathbf{L}_{\text{wall}}^{(g)}, \quad (21)$$

so that the transmission factors along individual rays are

$$\mathbf{A}^{(g)}(E) = \exp(-\mathbf{T}^{(g)}(E)), \quad (22)$$

with the exponential applied elementwise.

The mean transmission factor for geometry g at energy E is

$$\bar{A}(g, E) = \frac{1}{N_g} \sum_{i=1}^{N_g} A_i^{(g)}(E) = \left\langle e^{-T_i^{(g)}(E)} \right\rangle_i. \quad (23)$$

These mean transmissions summarise the effect of the voxel layout and walls for each detector geometry and energy and link the Monte Carlo model to the linear algebra formulation.

Linear algebra formulation and layout optimisation

For each detector geometry g and gamma ray line ℓ from nuclide ν with energy E_ℓ and emission probability $P_\gamma(\ell)$ the expected net peak count rate is

$$\hat{y}_{g\ell} = A_\nu P_\gamma(\ell) \varepsilon(E_\ell) \bar{A}(g, E_\ell), \quad (24)$$

where A_ν is the activity of nuclide ν in the cube, $\varepsilon(E_\ell)$ is the intrinsic full energy peak efficiency and $\bar{A}(g, E_\ell)$ is the mean transmission from the Monte Carlo model.

It is convenient to define a system factor

$$F(g, E_\ell) = \varepsilon(E_\ell) \bar{A}(g, E_\ell), \quad (25)$$

which collects detector efficiency and the effect of the voxel layout.

Let the detector geometries be labelled $g = 1, \dots, G$ and the selected lines be labelled $\ell = 1, \dots, L$. The measured net count rates $y_{g\ell}$ are stacked into a single vector y of length GL . The unknown activities are collected into

$$A = \begin{bmatrix} A_{\nu_1} \\ A_{\nu_2} \\ \vdots \\ A_{\nu_N} \end{bmatrix}, \quad (26)$$

with one component for each of the N nuclides in the model.

For a line ℓ that belongs to nuclide ν the corresponding row of the design matrix M is

$$M_{(g,\ell),\nu} = P_\gamma(\ell) F(g, E_\ell), \quad (27)$$

and the same row has zeros in all other columns. Collecting all rows gives the linear system

$$y \approx MA, \quad (28)$$

where M has GL rows and N columns.

For a fixed layout a weighted least squares estimate of the activities minimises

$$\chi^2(A) = (y - MA)^T W_y (y - MA), \quad (29)$$

where $W_y = \text{diag}(1/\sigma_{y_1}^2, \dots, 1/\sigma_{y_{GL}}^2)$ is built from the variances of the measured count rates. The minimiser is

$$A_{LS} = (M^T W_y M)^{-1} M^T W_y y. \quad (30)$$

Any negative components of A_{LS} are set to zero to enforce non negative activities.

The quality of the fit for a given layout is measured by the weighted sum of squared errors

$$\text{SSE} = (y - M A_{LS})^T W_y (y - M A_{LS}) = \sum_k \frac{(y_k - [M A_{LS}]_k)^2}{\sigma_{y_k}^2}, \quad (31)$$

where the index k runs over all measured peaks.

The voxel layout is represented by an integer label vector

$$\ell = \begin{bmatrix} \ell_1 \\ \ell_2 \\ \vdots \\ \ell_{27} \end{bmatrix}, \quad (32)$$

where $\ell_j \in \{0, 1, \dots, M-1\}$ selects one of M candidate materials (Cu, Al, PMMA) for voxel j .

Changing voxel j from material m_{old} to m_{new} changes its attenuation coefficient by

$$\Delta\mu(E) = \mu_{\text{new}}(E) - \mu_{\text{old}}(E). \quad (33)$$

For geometry g and ray i this updates the exponent and transmission as

$$T_i^{(g)}(E) \rightarrow T_i^{(g)}(E) + \Delta\mu(E) L_{ij}^{(g)}, \quad (34)$$

$$A_i^{(g)}(E) \rightarrow A_i^{(g)}(E) \exp(-\Delta\mu(E) L_{ij}^{(g)}). \quad (35)$$

Only the j th column of $L^{(g)}$ is needed for this update, so the mean transmissions $\bar{A}(g, E)$ and therefore the design matrix and SSE can be recomputed without rerunning the Monte Carlo.

The optimisation in the code uses greedy coordinate descent on ℓ with random restarts. At each step all candidate material changes for all voxels are tested and the one that gives the largest decrease in SSE is accepted. This is repeated until no more improvement is found or a maximum number of sweeps is reached. The best configuration is the one with the smallest SSE over all restarts.

Aluminium box density

The aluminium cube is modelled as a box of outer dimensions

$$L_{x,\text{out}}, \quad L_{y,\text{out}}, \quad L_{z,\text{out}} \quad (36)$$

and inner cavity dimensions

$$L_{x,\text{in}}, \quad L_{y,\text{in}}, \quad L_{z,\text{in}}. \quad (37)$$

The outer and inner volumes are

$$V_{\text{out}} = L_{x,\text{out}} L_{y,\text{out}} L_{z,\text{out}}, \quad V_{\text{in}} = L_{x,\text{in}} L_{y,\text{in}} L_{z,\text{in}}, \quad (38)$$

and the aluminium volume is

$$V_{\text{Al}} = V_{\text{out}} - V_{\text{in}}. \quad (39)$$

Given a measured mass m of the aluminium box the inferred density is

$$\rho_{\text{Al}} = \frac{m}{V_{\text{Al}}}. \quad (40)$$

Six face localisation model: uniform cube

For the six face localisation a single strong gamma line at energy E is used. For each face $f \in \{xp, xm, yp, ym, zp, zm\}$ a net count rate I_f is obtained by a small window integration. These are arranged into a vector of absolute intensities and normalised to give a relative pattern

$$I_{\text{meas},f} = \frac{I_f}{\sum_{f'} I_{f'}}. \quad (41)$$

In a simple fallback model the cube interior is treated as homogeneous with a single attenuation coefficient μ_{eff} and side length L . For a candidate source position (x, y, z) with $0 \leq x, y, z \leq L$ the path lengths to the six faces are

$$d_{xp} = L - x, \quad d_{xm} = x, \quad d_{yp} = L - y, \quad d_{ym} = y, \quad d_{zp} = L - z, \quad d_{zm} = z, \quad (42)$$

so that

$$I_f(x, y, z) \propto \exp(-\mu_{\text{eff}} d_f(x, y, z)). \quad (43)$$

These model intensities are normalised so that their sum is one, giving a model pattern $I_{\text{model},f}(x, y, z)$.

At each candidate position the discrepancy between data and model is quantified by the squared error

$$\text{SSE}(x, y, z) = \sum_f (I_{\text{meas},f} - I_{\text{model},f}(x, y, z))^2. \quad (44)$$

On a $3 \times 3 \times 3$ grid of voxel centres the code converts these SSE values into a probability like field by defining

$$p_{ijk} = \frac{1}{\text{SSE}_{ijk} + \varepsilon} \bigg/ \sum_{u,v,w} \frac{1}{\text{SSE}_{uvw} + \varepsilon}, \quad (45)$$

with a small stabiliser ε to avoid division by zero. By definition $\sum_{i,j,k} p_{ijk} = 1$. The voxel with the largest p_{ijk} is taken as the most likely source location and the full field provides a visual measure of localisation uncertainty.

Six face localisation model: heterogeneous cube

For a more realistic localisation the reconstructed material layout is used to construct a $3 \times 3 \times 3$ grid of attenuation coefficients

$$\mu_{ijk}(E) \quad \text{for } i, j, k \in \{0, 1, 2\} \quad (46)$$

at the localisation energy E . In practice the $\mu_{ijk}(E)$ are taken from the best fit configuration obtained in the layout optimisation described in the main text. The cube side is L and the voxel edge length is $\Delta = L/3$.

For a source position (x, y, z) the path towards a given face is approximated as a sequence of straight segments along one coordinate axis, crossing one voxel at a time. For example, for the $+x$ face the line integral is approximated as

$$\tau_{xp}(x, y, z) \approx \sum_m \mu_{i_m j_m k_m}(E) \Delta s_m, \quad (47)$$

where Δs_m is the path length within voxel (i_m, j_m, k_m) along the route from (x, y, z) to the plane $x = L$. The same construction is applied along $-x$, $+y$, $-y$, $+z$ and $-z$ to obtain τ_{xm} , τ_{yp} , τ_{ym} , τ_{zp} and τ_{zm} . Collecting these we define model intensities

$$I_{\text{model},f}(x, y, z) \propto \exp(-\tau_f(x, y, z)), \quad (48)$$

normalised so that the components sum to one. The SSE and probability map are then defined exactly as in the uniform case.

Localisation probability regions

The probability field p_{ijk} on the $3 \times 3 \times 3$ grid can be used to construct regions for the source position. To define, for example, the smallest region that contains 68 percent of the total probability:

1. Flatten the 27 values p_{ijk} into a list $\{p_n\}$ with $n = 1, \dots, 27$.
2. Sort this list in descending order to obtain $p_{(1)} \geq p_{(2)} \geq \dots \geq p_{(27)}$.
3. Form the cumulative sums

$$S_m = \sum_{n=1}^m p_{(n)}, \quad m = 1, \dots, 27. \quad (49)$$

4. Find the smallest m such that $S_m \geq 0.68$.

The corresponding set of m voxels defines the smallest discrete region that contains at least 68 percent of the total probability, and m is reported in the main text as the number of voxels in the 68 percent region. Exactly the same process can be used for 95 percent or other probability levels.

Nuclide identification from six faces

For each of the six spectra the code runs a peak finder that returns a set of peak energies $E_j^{(f)}$ on face f . These are compared to a line catalogue containing energies $E_\ell^{(\nu)}$ for each nuclide ν in a small library.

A catalogue line $E_\ell^{(\nu)}$ is considered matched to a measured peak $E_j^{(f)}$ if

$$|E_j^{(f)} - E_\ell^{(\nu)}| \leq \Delta E_{\text{tol}}, \quad (50)$$

with ΔE_{tol} a tolerance of order one keV. For each nuclide ν the number of matched lines n_ν and the mean absolute energy deviation

$$\overline{\Delta E}_\nu = \frac{1}{n_\nu} \sum_{\ell \in \text{matches for } \nu} |E_\ell^{\text{meas}} - E_\ell^{(\nu)}| \quad (51)$$

are computed, combining peaks from all six faces. The library is then ranked primarily by n_ν (more matched lines is better) and secondarily by $\overline{\Delta E}_\nu$ (smaller mean deviation is better) to obtain a simple nuclide identification list that is reported in the main text.

Error analysis

This section outlines how uncertainties propagate through the main stages of the analysis.

Uncertainty in efficiency calibration

The efficiency at E_i is

$$\varepsilon(E_i) = \frac{c_i}{A_{\text{Eu}} P_\gamma(E_i)}. \quad (52)$$

Assuming uncorrelated uncertainties in c_i , A_{Eu} and $P_\gamma(E_i)$, the variance is

$$\sigma_\varepsilon^2(E_i) = \left(\frac{\partial \varepsilon}{\partial c_i} \right)^2 \sigma_{c_i}^2 + \left(\frac{\partial \varepsilon}{\partial A_{\text{Eu}}} \right)^2 \sigma_{A_{\text{Eu}}}^2 + \left(\frac{\partial \varepsilon}{\partial P_\gamma} \right)^2 \sigma_{P_\gamma}^2, \quad (53)$$

with

$$\frac{\partial \varepsilon}{\partial c_i} = \frac{1}{A_{\text{Eu}} P_\gamma(E_i)}, \quad (54)$$

$$\frac{\partial \varepsilon}{\partial A_{\text{Eu}}} = -\frac{c_i}{A_{\text{Eu}}^2 P_\gamma(E_i)}, \quad (55)$$

$$\frac{\partial \varepsilon}{\partial P_\gamma} = -\frac{c_i}{A_{\text{Eu}} P_\gamma^2(E_i)}. \quad (56)$$

In log space:

$$\sigma_{\log_{10} \varepsilon} = \frac{1}{\ln 10} \frac{\sigma_\varepsilon}{\varepsilon}. \quad (57)$$

These $\sigma_{\log_{10} \varepsilon}$ enter the weights used in the linear fit for a and b .

Uncertainty in activity estimates

For the data vector y the variances are collected into

$$\text{Cov}(y) = \Sigma_y = \text{diag}(\sigma_{y_1}^2, \dots, \sigma_{y_{GL}}^2), \quad (58)$$

and the weight matrix $W_y = \Sigma_y^{-1}$ is used in the activity fit. Under linear model assumptions the covariance of the weighted least squares estimator is approximated by

$$\text{Cov}(A_{\text{LS}}) \approx (M^\top W_y M)^{-1}. \quad (59)$$

The diagonal elements give the variances $\sigma_{A_\nu}^2$ for each nuclide. In practice other systematic components from the efficiency model and attenuation coefficients are folded in by inflating the σ_{y_k} or by adding a relative scale uncertainty.

Uncertainty in aluminium density

The density $\rho_{\text{Al}} = m/V_{\text{Al}}$ depends on the mass m and on the outer and inner dimensions that define V_{Al} . Writing

$$V_{\text{out}} = L_{x,\text{out}} L_{y,\text{out}} L_{z,\text{out}}, \quad V_{\text{in}} = L_{x,\text{in}} L_{y,\text{in}} L_{z,\text{in}}, \quad V_{\text{Al}} = V_{\text{out}} - V_{\text{in}}, \quad (60)$$

and assuming independent uncertainties on m and each length, the derivatives are

$$\frac{\partial \rho_{\text{Al}}}{\partial m} = \frac{1}{V_{\text{Al}}}, \quad \frac{\partial \rho_{\text{Al}}}{\partial V_{\text{Al}}} = -\frac{m}{V_{\text{Al}}^2}, \quad (61)$$

and

$$\frac{\partial V_{\text{out}}}{\partial L_{x,\text{out}}} = L_{y,\text{out}} L_{z,\text{out}}, \quad \frac{\partial V_{\text{out}}}{\partial L_{y,\text{out}}} = L_{x,\text{out}} L_{z,\text{out}}, \quad \frac{\partial V_{\text{out}}}{\partial L_{z,\text{out}}} = L_{x,\text{out}} L_{y,\text{out}}, \quad (62)$$

$$\frac{\partial V_{\text{in}}}{\partial L_{x,\text{in}}} = L_{y,\text{in}} L_{z,\text{in}}, \quad \frac{\partial V_{\text{in}}}{\partial L_{y,\text{in}}} = L_{x,\text{in}} L_{z,\text{in}}, \quad \frac{\partial V_{\text{in}}}{\partial L_{z,\text{in}}} = L_{x,\text{in}} L_{y,\text{in}}. \quad (63)$$

Since $V_{\text{Al}} = V_{\text{out}} - V_{\text{in}}$ the chain rule gives

$$\frac{\partial \rho_{\text{Al}}}{\partial L_k} = \frac{\partial \rho_{\text{Al}}}{\partial V_{\text{Al}}} \frac{\partial V_{\text{Al}}}{\partial L_k} \quad (64)$$

for each length parameter L_k . If σ_m is the uncertainty on the mass and σ_{L_k} the uncertainty on each dimension, the variance in ρ_{Al} is approximated by

$$\sigma_\rho^2 \approx \left(\frac{\partial \rho_{\text{Al}}}{\partial m} \right)^2 \sigma_m^2 + \sum_k \left(\frac{\partial \rho_{\text{Al}}}{\partial L_k} \right)^2 \sigma_{L_k}^2. \quad (65)$$

Uncertainty in layout reconstruction

A simple way to characterise strength of the model is to evaluate how much the SSE increases when individual voxels are forced to alternative materials or when random restarts converge to different layouts.

References for mathematical methods and algorithms

Energy calibration, efficiency and resolution in gamma spectrometry. The quadratic energy calibration, power law or polynomial efficiency models and Gaussian peak fitting with FWHM² are standard in gamma ray spectrometry. Can be found in G. F. Knoll, *Radiation Detection and Measurement*, 4th ed. (Wiley, 2010), especially Chapters 7–8, and G. Gilmore, *Practical Gamma Ray Spectrometry*, 2nd ed. (Wiley, 2008), Chapters 4, 5 and 8. Additional background on detector response and resolution modelling is given in W. R. Leo, *Techniques for Nuclear and Particle Physics Experiments*, 2nd ed. (Springer, 1994).

Weighted least squares, covariance propagation and activity fitting. The weighted least squares estimator, its covariance $\text{Cov}(A_{\text{LS}}) \approx (M^T W_y M)^{-1}$ and Gaussian error propagation for functions such as $E(c)$ and $\varepsilon(E)$ follow in standard linear algebra. See, for example, P. R. Bevington and D. K. Robinson, *Data Reduction and Error Analysis for the Physical Sciences*, 3rd ed. (McGraw Hill, 2003), and W. H. Press *et al.*, *Numerical Recipes: The Art of Scientific Computing*, 3rd ed. (Cambridge University Press, 2007), Chapters 15–16.

Exponential attenuation and XCOM based coefficients. The use of $I = I_0 \exp[-(\mu/\rho)x]$ and interpolation of mass attenuation coefficients is standard in radiation transport. The NIST XCOM database is described in M. J. Berger *et al.*, “XCOM: Photon Cross Sections Database,” NIST Standard Reference Database 8 (XGAM). Discussions of exponential attenuation and build up can be found in Knoll (Chapter 2) and Leo (Chapter 2).

Monte Carlo ray tracing and voxel based attenuation. The voxel based Monte Carlo model which records path lengths $L_{ij}^{(g)}$ and forms transmission factors $\exp[-L^{(g)}\mu(E)]$ is closely related to track length estimators in radiation transport Monte Carlo. Similar techniques found in: M. H. Kalos and P. A. Whitlock, *Monte Carlo Methods*, 2nd ed. (Wiley VCH, 2008), and K. K. Shultis and R. E. Faw, *Radiation Shielding* (American Nuclear Society, 2000). The three dimensional digital differential analyser (DDA) style ray marching used to step through voxels follows the algorithm of J. Amanatides and A. Woo, “A fast voxel traversal algorithm for ray tracing,” in *Eurographics '87*, pp. 3–10.

Greedy layout optimisation and coordinate descent. The discrete layout optimisation, where voxel materials are updated to reduce a model to measure discrepancy (the SSE), is an example of greedy coordinate descent. General overviews of coordinate descent and local search methods can be found in J. Nocedal and S. J. Wright, *Numerical Optimization*, 2nd ed. (Springer, 2006), Chapter 6, while non negative least squares and constrained activity fits are discussed in C. L. Lawson and R. J. Hanson, *Solving Least Squares Problems* (SIAM, 1995).

Six face localisation and probability maps. The six face localisation model treats the relative face intensities as an attenuation dependent forward model with a misfit function. The conversion of the misfit field into a discrete probability map p_{ijk} and the construction of smallest regions containing a given probability mass follow from Bayesian credible regions. See D. S. Sivia and J. Skilling, *Data Analysis: A Bayesian Tutorial*, 2nd ed. (Oxford University Press, 2006), and G. Cowan, *Statistical Data Analysis* (Oxford University Press, 1998), Chapters 2–4.

Peak finding and nuclide identification. The peak finding and catalogue matching used for nuclide identification follows standard practice in gamma spectrometry software: peaks are located in each spectrum, matched to tabulated line energies within a tolerance and scored by the number of matches and mean deviation. Descriptions of such algorithms and their practical implementation are given in G. Gilmore, *Practical Gamma Ray Spectrometry*, 2nd ed. (Wiley, 2008), Chapters 6–7, and in the documentation of common analysis packages such as Genie 2000 and FitzPeaks (see manufacturer manuals for implementation details).

Appendix B: Gamma Spectroscopy Data and Example Spectra

1 Data organisation

This appendix provides an overview of the data structure and files used in the analysis. The data are organised into several directories, each containing relevant spectra files for gamma spectrometry, attenuation measurements and calibration data.

1.1 Directory structure

The raw data are stored under the path

```
C:\Users\sgmturn2\OneDrive - The University of Liverpool\Documents\University\Year
3\Semester 1\Practical Physics III\Section C\Data
```

For the local Python analysis the data were copied into an **Analysis** folder. The folder layout under

```
C:\Users\marcu\OneDrive - The University of Liverpool\Documents\University\Year
3\Semester 1\Practical Physics III\Section C\Analysis\Data
```

is:

Attenuation Data

```
Face 1 (subfolders 1:9)
Face 2 (subfolders 1:9)
Face 4 (subfolders 1:9)
```

Gamma Spectroscopy

```
Face 1
Face 2
Face 4
Face 5
Face 6
```

The **Attenuation Data** folder contains Cs 137 attenuation runs for multiple cube positions and faces, while the **Gamma Spectroscopy** folder contains HPGe cube spectra and Eu 152 calibration data, as described below.

1.2 Attenuation Data directory

This directory contains data related to gamma attenuation measurements through various materials, organised by different faces of the cube setup.

- **Face 1, Face 2, Face 4:** each of these directories contains multiple subdirectories corresponding to the different beam positions on that face, labelled 1 to 9.

- Files such as `CsAluminiumBox.ava` and `CsNoAttenuation.ava` represent specific attenuation data related to Cs 137 sources used for calibration and testing.
- The subdirectories labelled 1, 2, 3 and so on correspond to different measurement configurations for Face 1, Face 2 and Face 4.

1.3 Gamma Spectroscopy directory

This directory contains the gamma spectrometry data, organised by the faces of the cube setup, which were used to identify radionuclides and analyse their activity.

- Each **Face X** subdirectory contains `.ava` and `.csv` files. The `.ava` files contain the raw spectrometric data, while the `.csv` files store the processed data and are used in the analysis.
- Files such as `EuCalibrationQ1.ava` and `EuCalibrationQ1.csv` correspond to calibration spectra for the Eu 152 source, which was used for energy calibration, efficiency estimation and resolution checks.

2 HPGe calibration and example spectra

2.1 Eu-152 lines and calibration quantities

The Eu 152 calibration spectrum was used to define the HPGe energy scale, full energy peak efficiency and resolution. Peak centroids were obtained with a peak finder followed by Gaussian fits to the known Eu 152 lines. For the efficiency calibration, a set of Eu lines with known emission probabilities $P_\gamma(E_i)$ were selected from nuclear data tables [1]. For each line the net peak area N_i was found by integrating the peak with sideband background subtraction, and converted to a count rate $c_i = N_i/t_{\text{live}}$. The absolute full energy peak efficiency at energy E_i is then

$$\varepsilon(E_i) = \frac{c_i}{A_{\text{Eu}}P_\gamma(E_i)}, \quad (1)$$

with the fitted power law model

$$\log_{10} \varepsilon(E) = a + b \log_{10} \left(\frac{E}{\text{keV}} \right) \quad (2)$$

documented in Appendix A.

Table 1 summarises a subset of the Eu 152 lines used in the calibration. Numerical values in red are taken from the Python analysis and should be filled once the final run is available.

Table 1: Selected Eu 152 lines used for HPGe calibration. Tabulated energies and emission probabilities P_γ are taken from NuDat 3.0 [1]. Net peak areas N_i are from Gaussian fits to the calibration spectrum, with uncertainties from counting statistics. Efficiencies are obtained from the calibration formula with propagated uncertainties.

E_{tab} (keV)	P_γ (%)	N_i (counts)	$\varepsilon(E_i)$	$E_{\text{fit}} - E_{\text{tab}}$ (keV)
121.8	28.50	1.36×10^4	3.63	-1.387
244.7	7.60	2.01×10^3	2.01	-1.408
344.3	26.70	4.74×10^3	1.35	-1.215
778.9	12.90	9.16×10^2	5.39×10^{-1}	-1.213
964.1	14.60	8.12×10^2	4.22×10^{-1}	-1.266
1408.0	21.00	7.85×10^2	2.84×10^{-1}	-1.238

2.2 Energy resolution

The energy resolution was characterised by fitting Gaussians to the same Eu lines. From the fitted standard deviations σ_E the full width at half maximum values were

$$\text{FWHM}(E_i) = 2.3548 \sigma_E. \quad (3)$$

A simple model was adopted in which the squared resolution depends linearly on energy,

$$\text{FWHM}^2(E) = a_{\text{res}} + b_{\text{res}} E, \quad (4)$$

with best fit parameters

$$a_{\text{res}} = 1.166 \text{ keV}^2, \quad b_{\text{res}} = 3.1 \times 10^{-3} \text{ keV}, \quad \chi^2/\text{NDF} = 11.9/4 \approx 3.0. \quad (5)$$

The fitted curve and residuals are summarised in Fig. [1](#).

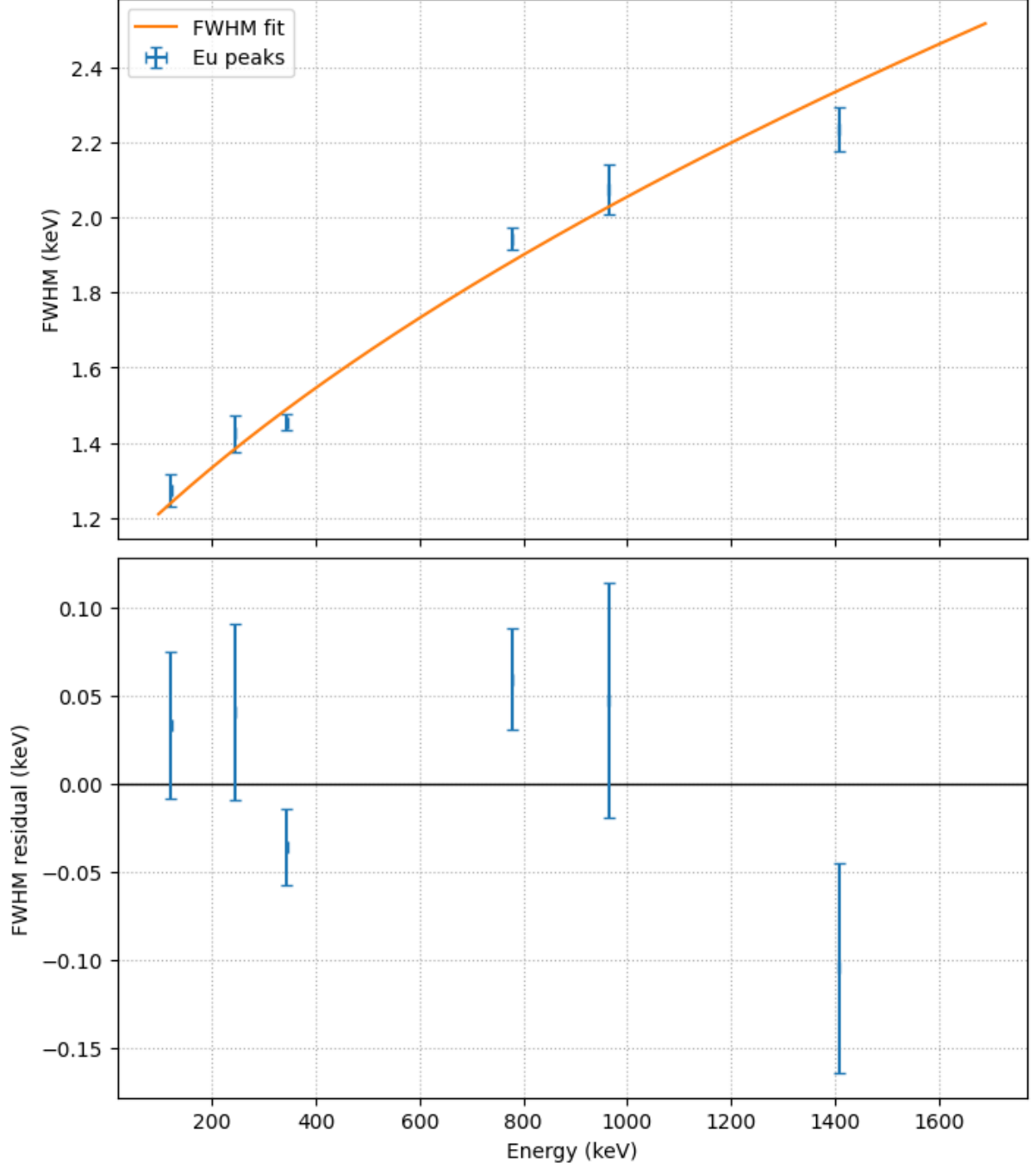


Figure 1: Energy resolution from Eu 152 calibration. Top panel: FWHM as a function of energy with the fitted model $\text{FWHM}^2(E) = a_{\text{res}} + b_{\text{res}}E$ overlaid. Bottom panel: residuals. The best fit parameters are $a_{\text{res}} = 1.166 \text{ keV}^2$ and $b_{\text{res}} = 3.1 \times 10^{-3} \text{ keV}$, with $\chi^2/\text{NDF} = 11.9/4 \approx 3.0$. The trend is consistent with a combination of an energy independent term and a contribution that grows with increasing energy.

2.3 Efficiency curve

The efficiency calibration and its residuals are shown in Fig. 2. The fit parameters are those reported in Appendix A.

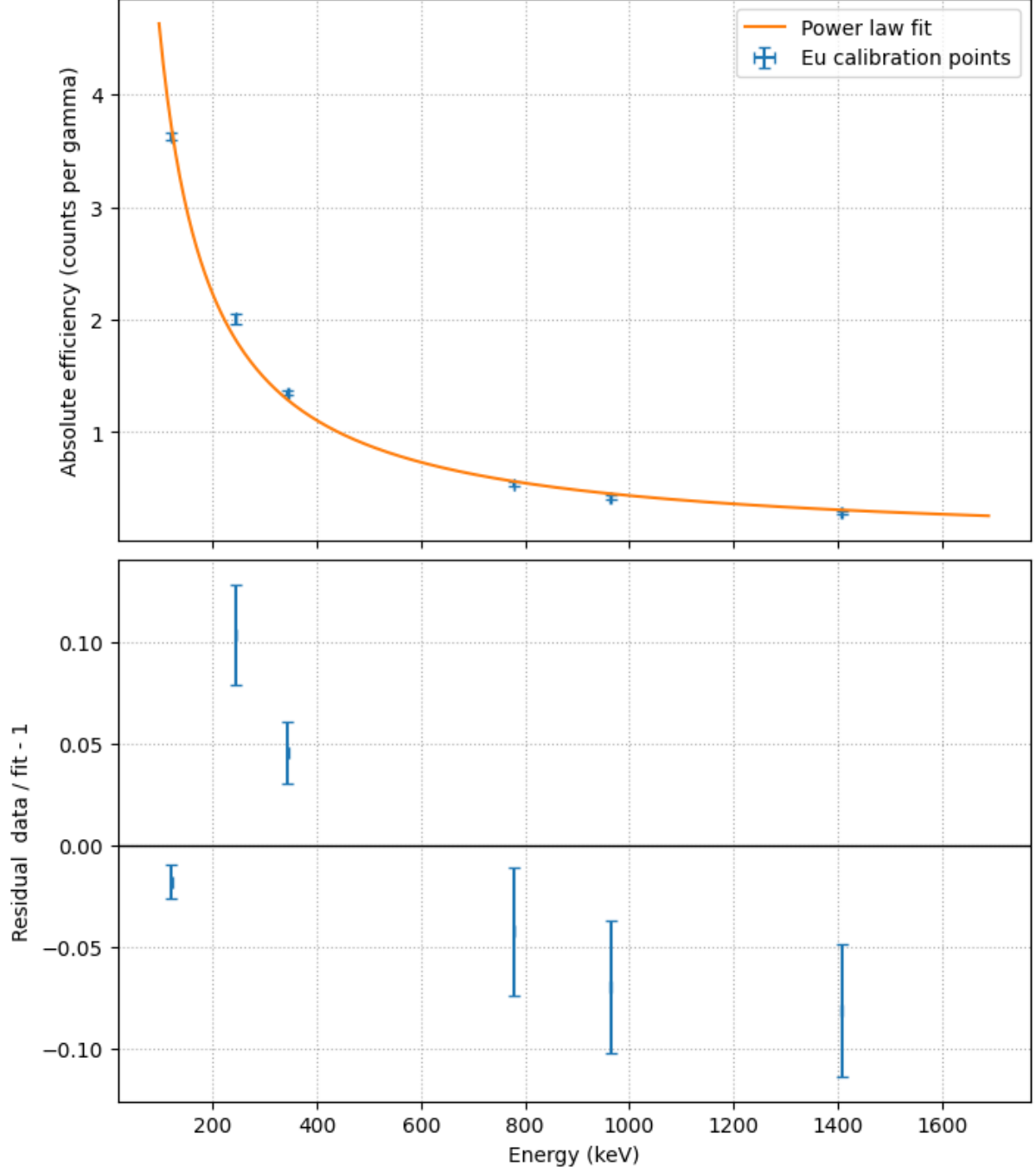


Figure 2: Detector full energy peak efficiency from Eu 152 calibration (top) with the power law fit $\log_{10} \varepsilon(E) = a + b \log_{10}(E/\text{keV})$ overlaid, and fractional residuals $(\varepsilon_{\text{data}}/\varepsilon_{\text{fit}} - 1)$ of the efficiency fit (bottom). The best fit parameters are $a = 2.682$ and $b = -1.014$, with $\chi^2/\text{NDF} = 45.2/4 \approx 11.3$. The several percent structure in the residuals is treated as a systematic uncertainty on activity estimates.

2.4 Cs-137 calibration spectrum

The NaI(Tl) calibration spectrum used to check gain and resolution and to provide an indicative efficiency point at 661.7 keV is shown in Fig. 3.

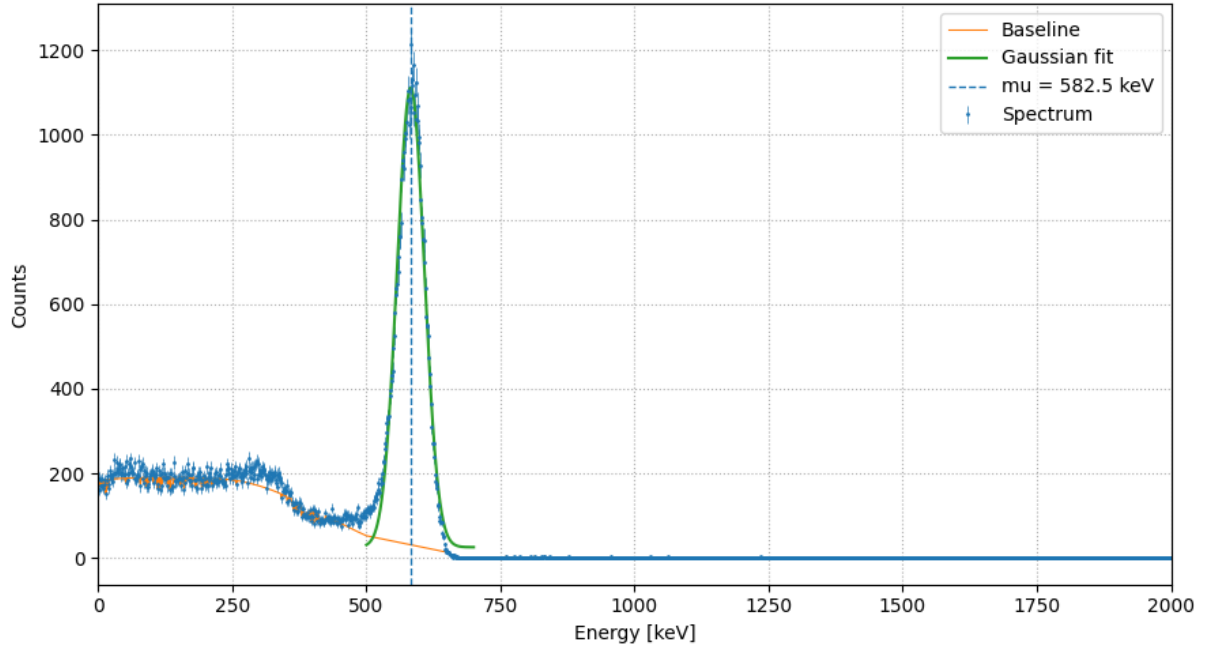


Figure 3: Cs 137 spectrum from a 2×2 inch NaI(Tl) scintillation detector (Mirion Model 802 series) in the no attenuation configuration. A Gaussian fit to the main peak is shown with the fitted centroid and full width at half maximum (FWHM). The centroid is around 582.5 keV with FWHM about 59.7 keV, consistent with a lower resolution detector and a different gain setting. This spectrum is used as a qualitative check on gain and resolution and for an indicative efficiency point at 661.7 keV.

2.5 Gamma Spectroscopy cube spectra

The cube spectra used for radionuclide identification and activity fitting are illustrated in Fig. 4.

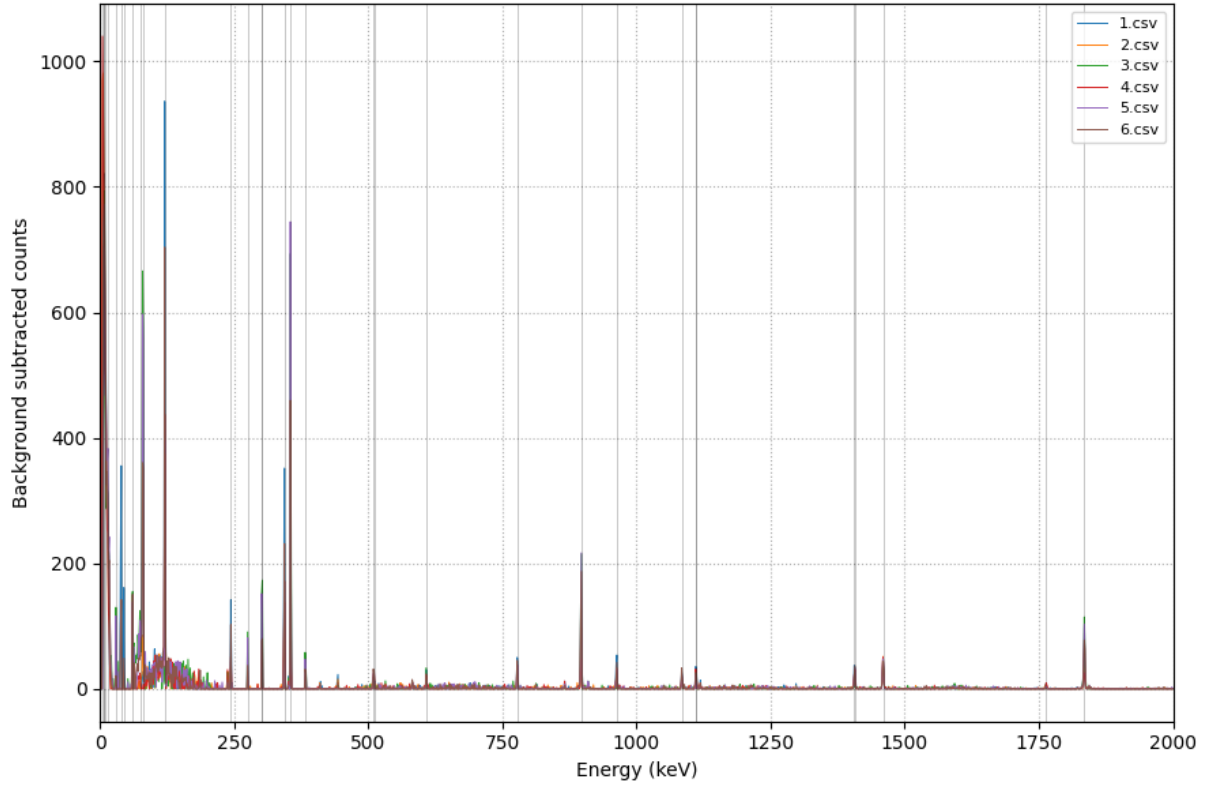


Figure 4: Background subtracted spectra from the Gamma Spectroscopy folder for the six cube positions. Vertical lines mark detected peaks used in radionuclide identification. Eu 152 lines dominate the spectra, with additional matches to Y 88, Ba 133 and weaker Na 22 lines also present.

2.6 Attenuation Data spectra

The Cs 137 attenuation spectra used to constrain aluminium density and attenuation coefficients for the voxel model are shown in Fig. 5.

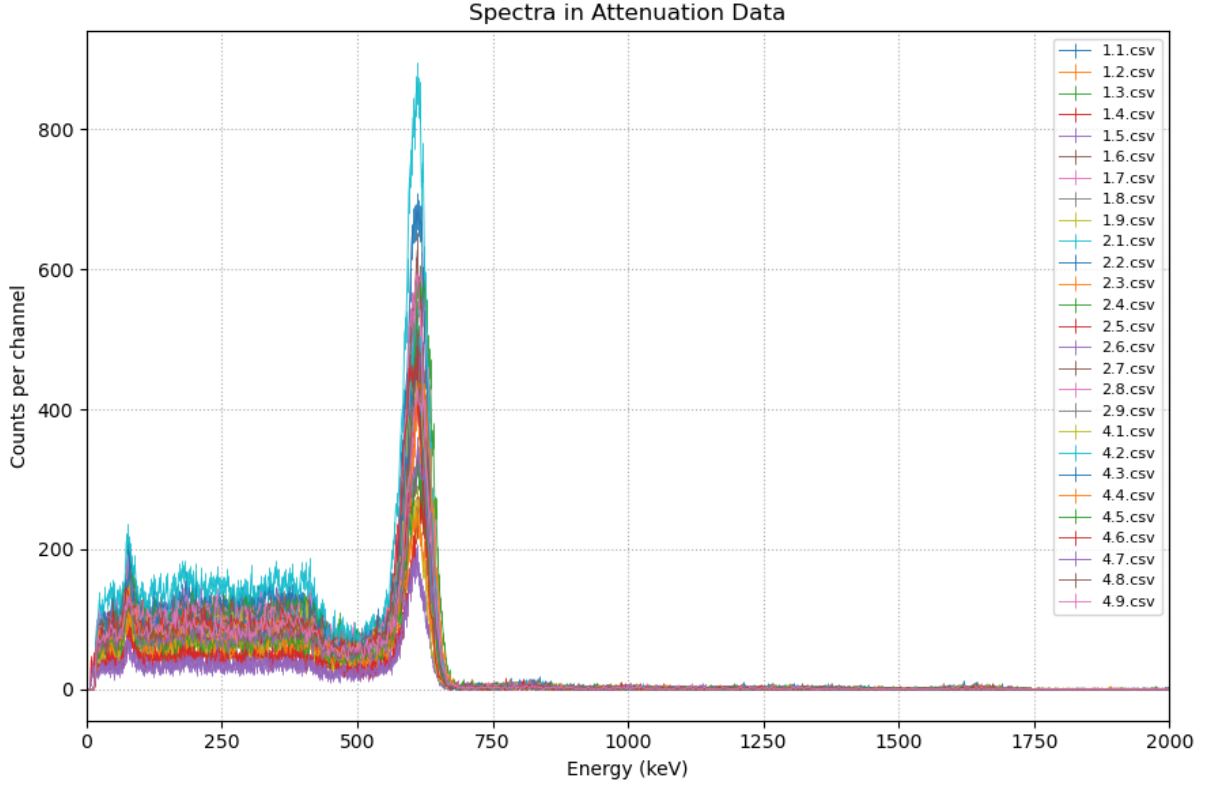


Figure 5: Cs 137 spectra from the Attenuation Data set for multiple cube positions. Variations in peak height and low energy continuum illustrate differing attenuation paths through the cube materials. These spectra are used to cross check the aluminium density and the attenuation coefficients employed in the voxel model and layout optimisation.

3 Sample Gamma Spectroscopy data

Below is a sample of the gamma spectrometry data from a CSV file. The full dataset is available in the associated data files and was read directly by the Python analysis scripts.

Channel	Energy (keV)	Net Counts
1	6.64	821
2	39.06	355
3	44.05	161
4	60.26	134
5	120.12	936
6	243.58	142
7	343.34	351
8	354.57	104
9	607.72	30
10	777.32	49

4 Detector square layout

Figure 6 shows the layout of the detector squares, with faces 1, 2 and 4 numbered. The left image displays the detector squares with their respective numbers, while the right image shows

the face numbers only. These numbers represent the face of the gamma cube corresponding to each square in the setup and are used to map attenuation measurements onto the voxel model.

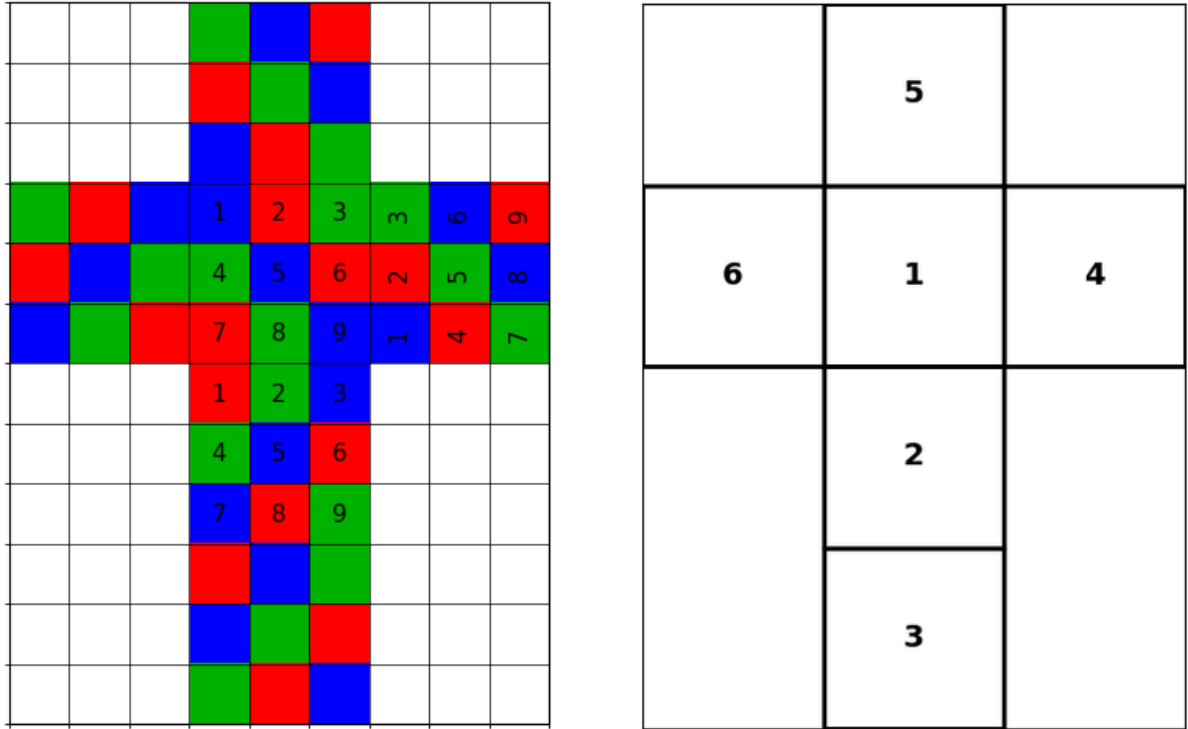


Figure 6: Detector square layout, showing faces 1, 2 and 4 numbered (left) and face numbers only (right).

5 Detector specifications

5.1 HPGe detector

The cube spectra and Eu 152 calibration data were acquired with a Canberra HPGe detector of type BE3825, mounted in a vertical Slimline cryostat and equipped with an iPA II intelligent preamplifier [2]. The BEGe family is designed to provide good energy resolution from a few keV up to several MeV, with a relatively thin, wide crystal that improves efficiency for typical sample geometries.

For the BE3825 model the active germanium area is about 38 cm^2 with a thickness of 25 mm, giving a typical relative efficiency near 26 % at 1332 keV [2]. The detector uses a carbon composite entrance window that maintains low energy efficiency down to a few keV while remaining mechanically robust, and a flat, non bulletized crystal that maintains high efficiency for samples placed close to the end cap.

The manufacturer specified FWHM resolutions for BEGe detectors of this size are around 0.7 keV at 122 keV and about 1.9 keV at 1332 keV when used with a suitable digital MCA [2]. These values are consistent with the empirical resolution model obtained from the Eu 152 calibration in this work and justify the use of narrow integration windows for line based analysis.

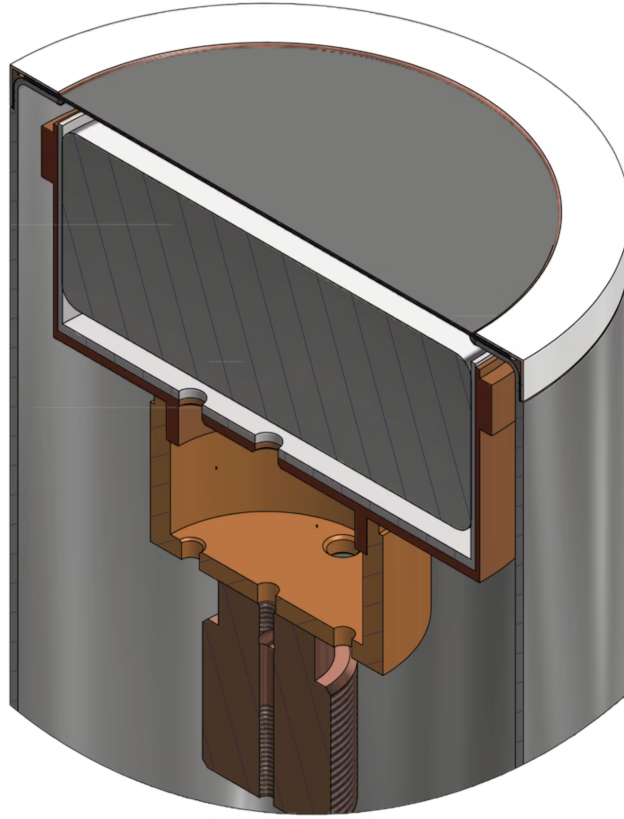


Figure 7: Photograph of a Mirion BEGe type HPGe detector similar to the BE3825 unit used for the cube spectroscopy measurements, showing the vertical cryostat, end cap and approximate detector geometry.

5.2 NaI(Tl) scintillation detector

The attenuation measurements were performed with a Mirion Model 802 NaI(Tl) scintillation detector with a 2×2 inch cylindrical crystal [3]. The detector is supplied as a hermetically sealed assembly that integrates the NaI(Tl) crystal, a photomultiplier tube, an internal magnetic and light shield and an aluminium housing with a 14 pin connector.

For the 2×2 inch configuration the guaranteed resolution at the 662 keV Cs 137 line is of order 8.5 % FWHM [3]. This is consistent with the broad peak observed in Fig. 3 and reflects the lower intrinsic resolution of scintillators compared to HPGe detectors. The aluminium entrance window has a nominal thickness of 0.5 mm, and the assembly includes an internal reflector layer and a conetic magnetic shield that improve light collection and stability.

The 802 series detectors are typically operated at a cathode to anode high voltage close to +1100 V, with dynode chain voltages chosen to balance gain and linearity [3]. In this experiment the detector was coupled to a digital tube base and multi channel analyser that provided the gain, shaping and live time information used to extract the Cs 137 peak areas and attenuation curves.

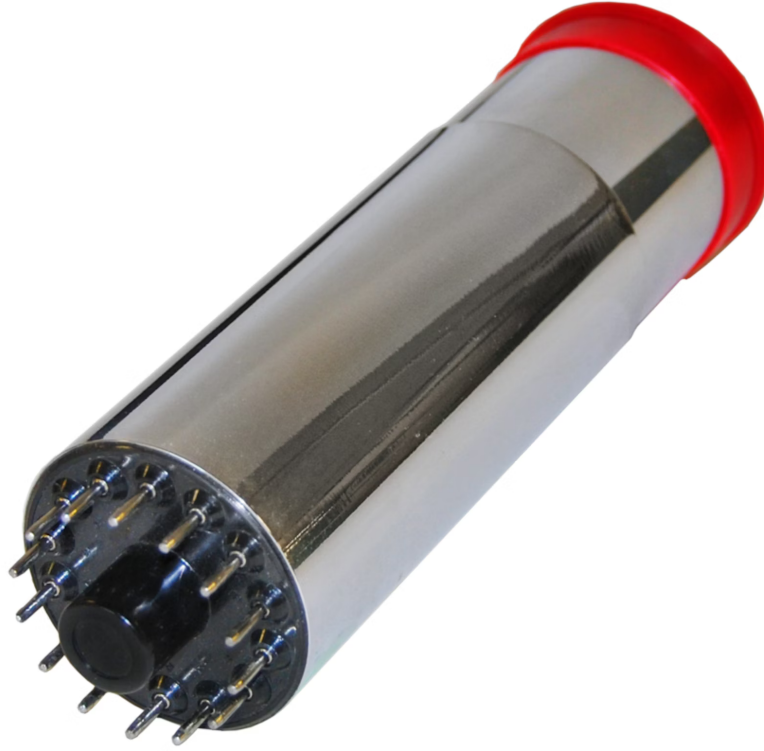


Figure 8: Photograph of a Mirion 802 series 2×2 inch NaI(Tl) scintillation detector similar to the unit used for the attenuation measurements, showing the cylindrical crystal housing and connector.

6 Appendix summary

The Gamma Spectroscopy data provide net counts at various energies for each detector position, while the Attenuation Data show how counts decrease as gamma rays pass through different material paths. Together with the detector layout, the Eu 152 calibration results, the six face localisation analysis and the HPGe and NaI(Tl) detector specifications, these files underpin the modelling and reconstruction described in the main report.

References

- [1] National Nuclear Data Center, “NuDat 3.0 Nuclear Structure and Decay Data,” Brookhaven National Laboratory (accessed 2024).
- [2] Mirion Technologies, “BEGe Broad Energy Germanium Detectors” specification sheet SPC 134 EN C (2025).
- [3] Mirion Technologies, “802 Scintillation Detectors” datasheet CSP0232 (2017).